Assignment

Part A : Practice Tasks :

Q1

```
var Name="Harsh";
var birthyear = 2004;
var learningstatus = true;
var favpl = ['python','C','java','C++','javascript'];
var profiled = {
    name:"harsh",
    age:21,
    hobby:"coding",
    learningstatus:true
}
console.log(Name,typeof(Name));
console.log(birthyear,typeof(birthyear));
console.log(learningstatus,typeof(learningstatus));
console.log(favpl,typeof(favpl));
console.log(profiled,typeof(profiled));
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART A> node Q1.js
Harsh string
2004 number
true boolean
[ 'python', 'C', 'java', 'C++', 'javascript' ] object
{ name: 'harsh', age: 21, hobby: 'coding', learningstatus: true } object
PS C:\Users\HomePC\Desktop\Mini project\PART A>
```

Q2

```
var products = [
        {
```

```
            id:1,
            name:"mobile",
            price:10000,
            inStock:true
           },
           {
             id:2,
             name:"TV",
             price:20000,
             inStock:false
           },
           {
             id:3,
             name:"Refrigerator",
             price:25000,
             inStock:true
           },
           {
             id:4,
             name:"AC",
             price:25000,
             inStock:false
           }
          ];

products.push({
    id:5,
    name:"Headphone",
    price:900,
    inStock:true
});
console.log("Headphone is pushed successfully into array");
```

```
console.log("item at index 1 is removed from the products array");
products.splice(1,1);

products[0].price=13000;
console.log("Updated price of mobile:"+products[0].price);

console.log("Products:");
products.forEach(p=>console.log(p.name));

console.log("Products in stock are:")
products.filter(product=>product.inStock).forEach(product=>console.log(product.name));
```

```
Headphone is pushed successfully into array
item at index 1 is removed from the products array
Updated price of mobile:13000
Products:
mobile
Refrigerator
AC
Headphone
Products in stock are:
mobile
Refrigerator
Headphone
```

Q3

```
let num = 18
if(num>0){
   console.log("THE NUMBER IS POSITIVE");
}
else if(num == 0){
   console.log("THE NUMBER IS ZERO");
}
else{
   console.log("THE NUMBER IS NEGATIVE");
}
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART A> node Q3.js
THE NUMBER IS POSITIVE
```

Q4

```
console.log("Numbers from 1 to 20 are:")
for(let i=1 ; i<=20 ; i++){
   console.log(i+"\n");
}

console.log("Even numbers from 1 to 20 are:")
for(let i=1 ; i<=20 ; i++){
   if(i%2==0){
            console.log(i+"\n");
         }
}

console.log("Numbers from 10 to 1 are:");
for(let i=10 ;i>=1;i--){
   console.log(i);
}
```

```
Numbers from 1 to 20 are:
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Even numbers from 1 to 20 are:
2
4
6
8
10
12
14
16
18
20
```

```
Numbers from 10 to 1 are:
10
9
8
7
6
5
4
3
2
1
```

Q5

```
function celsiusToFahrenheit(){
    let c = 35;
    return (c*9/5)+32;
}
console.log("The temperature in fahrenheit is:"+celsiusToFahrenheit());

function EvenOdd(){
    let n=5;
    if(n%2==0){
        return "Number is Even";
    }
    else{
        return "Number is Odd";
    }
}
console.log(EvenOdd());

function LargestNum(a,b){
    var a = 5;
    var b = 4;
    return a > b ? a : b;
}
console.log("The largest number is:"+LargestNum());
```

```
The temperature in fahrenheit is:95
Number is Odd
The largest number is:5
```

PART : B

Q1

```
function countvowels(){
    let str = "Hello World";
    let count=0;
    let vowels= "aeiouAEIOU";

    for(let char of str){
        if(vowels.includes(char)){
            count++;
        }
    }
    return count++;
}
console.log("Number of Vowels:"+countvowels());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q1.js
Number of Vowels:3
```

Q2

```
function reverseArray(){
    let arr=[5,6,7,8];
    let res = [];

    for(let i=arr.length-1;i>=0;i--){
        res.push(arr[i]);
    }
    return res;
}
console.log("Reversed array is:"+"["+reverseArray()+"]");
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q2.js
Reversed array is:[8,7,6,5]
```

Q3

```
function secondLargest(){
    let arr = [7,18,45,100];
    let largest = -Infinity;
    let second = -Infinity;

    for(let num of arr){
        if(num>largest){
            second = largest;
            largest=num;
        }
        else if(num>second && num!==largest){
            second=num;
        }
    }
    return second;
}
console.log("Second largest number is:"+secondLargest());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q3.js
Second largest number is:45
```

Q4

```
function removeDuplicates(){
    let unique=[];
    let arr = [1,2,3,3,4];
    for(let item of arr){
        if(!unique.includes(item)){
            unique.push(item);
        }
    }
    return unique;
}
```

```
console.log("array after removing duplicates:"+removeDuplicates());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q4.js
array after removing duplicates:1,2,3,4
```

Q5

```
function isPalindrome(){
    let str = "madam";
    let reversed = str.split("").reverse().join("");
    return str === reversed;
}
console.log("The Array is palindrome:"+isPalindrome());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q5.js
The Array is palindrome:true
```

Q6

```
function longestUniqueSubstring(){
    let maxLength = 0;
    let str = "Javascript";
    for(let i=0;i<str.length;i++){
        let seen="";
        for(let j=i;j<str.length;j++){
            if(seen.includes(str[j]))
                break;
            seen+=str[j]
            maxLength=Math.max(maxLength,seen.length);
        }
    }
    return maxLength;
}
console.log("The length of the longest substring without repeating
characters:"+longestUniqueSubstring());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q6.js
The length of the longest substring without repeating characters:8
```

Q7

```
function mergeSortedArrays(){
    let arr1 = [2,3,5];
    let arr2 = [1,4,6];
    let result = [];
    let i=0,j=0;

    while(i<arr1.length && j<arr2.length){
        if(arr1[i]<arr2[j]){
            result.push(arr1[i++]);
        }
        else{
            result.push(arr2[j++]);
        }
    }
    return result.concat(arr1.slice(i)).concat(arr2.slice(j));
}
console.log("The sorted array is:"+"["+mergeSortedArrays()+"]");
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q7.js
The sorted array is:[1,2,3,4,5,6]
```

Q8

```
function longestCommonPrefix(){
    let arr=["WorldWide","World War","World Wide Web"];
    if(arr.length===0){
        return "";
    }
    let prefix = arr[0];
    for(let i=1;i<arr.length;i++){
        while(!arr[i].startsWith(prefix)){
            prefix = prefix.slice(0,-1);
            if(prefix===""){
```

```
            return "";
        }
    }
  }
  return prefix;
}
console.log("The longest common prefix is:"+longestCommonPrefix());
```

```
PS C:\Users\HomePC\Desktop\Mini project\PART B> node Q8.js
The longest common prefix is:World
```

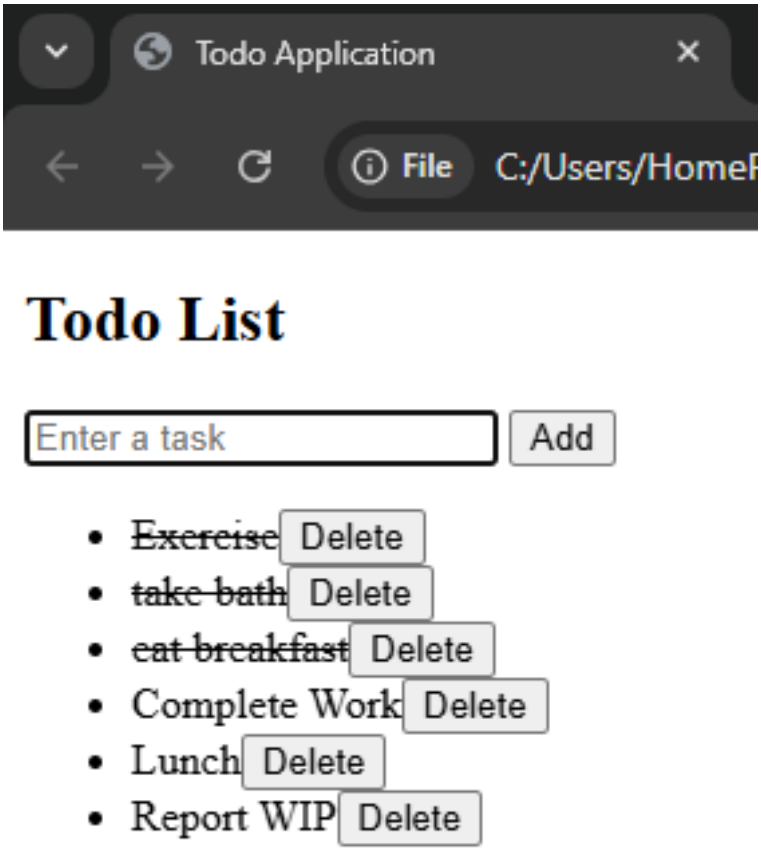
PART : C

Q2 : Todo Application

Html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Todo Application</title>
</head>
<body>
  <script src="miniproject.js"></script>
  <h2>Todo List</h2>
  <input type="text" id="ui" placeholder="Enter a task">
  <button onclick="addTask()">Add Task</button>
  <ul id="taskList"></ul>
</body></html>
```

Javascript

```
function addTask(){
    let task = document.getElementById("ui").value;

    if(task==="")
        return ;

    let li = document.createElement("li");

    let span = document.createElement("span");
    span.textContent = task;

    span.onclick = function(){
        span.style.textDecoration = "line-through";
    };

    let btn = document.createElement("button");
    btn.textContent = "Delete";

    btn.onclick = function(){
        li.remove();
    };

    li.appendChild(span);
    li.appendChild(btn);
    document.getElementById("taskList").appendChild(li);
```

```
    document.getElementById("ui").value="";
}
```